Efficient Calibration of Industrial Robots in Multi-Spherical Constraints Supplementary Material

Anonymous

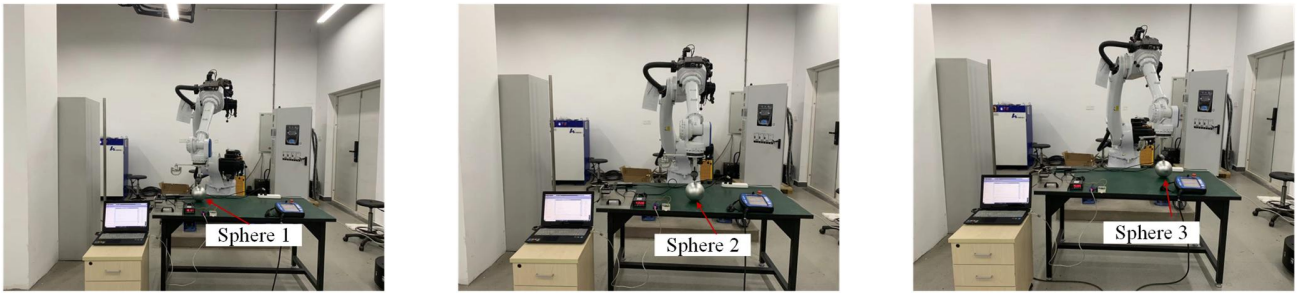
This supplementary file includes a table and several figures related to the symbols and data collection process, as well as the complete convergence proof of the MSCA algorithm.

I. Additional Tables

TABLE S.I. Symbol list.

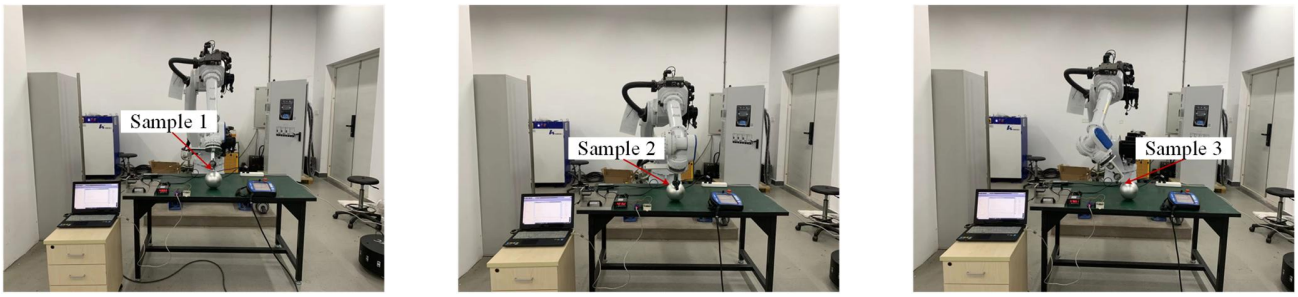
Symbol	Explanation
Υ_i	Transformation matrix.
a, α, θ, d	Link length, link twist angle, link offset, joint angle.
B, S	Rotation matrix, position vector.
$d\Upsilon$	Holomorphic differential Υ.
ΔE	Kinematic error.
H	Jacobian matrix of the kinematic error model.
ΔO	Kinematic parameters error vector.
$\Delta\alpha$, Δa, $\Delta\theta$, Δd	Vector of the kinematic parameter deviations.
L	Loss function.
$\hat{V}$, V	Theoretical and measured draw-wire lengths.
Z	Fixed point on the ground.
$S(O)$	Theoretical robot end-effector position
N	Sample count (measurement configurations).
O	D-H parameter vector.
Q	Covariance matrix of system noise.
v_k	Measurement noise.
K	Kalman filter gain matrix.
M	Measurement noise covariance matrix
I	Unit matrix.
μ	Regularization parameter.
a_6, a'_6	Initial and defined value of the sixth axis link length.
Δb	Difference between the length of the dial gauge body and its real-time reading.
Φ	Spherical constraint equation.
S_i^u	The i-th position on the u-th sphere.
ξ^u	The u-th sphere's center.
R_u	The radius of the u-th sphere.
U	Total number of spheres.
$<\cdot>$	The inner product between two matrices
κ_u	Constraint coefficient of sphere u.
γ^u	Lagrange multiplier of sphere u.
η^u	The coefficient of the Lagrange multiplier for the u-th sphere.
ε_1, ε_2, ε_3	Custom functions.
$\sigma_{1,i}$, $\sigma_{2,i}$, $\sigma_{3,i}$, $\sigma_{4,i}$	Custom functions.
O_0, Z_0, γ_0	Initial values of O, Z, and γ.
T	Maximum iteration per time step in the IEKF step.
C_1	Maximum iteration to Z in the MSCA step.
C_2	Maximum iteration for updating ξ^u, O and γ^u in the MSCA step.
$\{q_{i,1}, \ldots, q_{i,6}\}$	Six robotic rotation angles.

II. Additional Figures



(a) (b) (c)

.Fig. S.1. The data collection process for different spheres (D1-3). Spheres 1-3 correspond to datasets D1-3.



(a) (b) (c)

Fig. S.2. The data collection process at different positions on a sphere. Samples 1-3 represent three different positions on the sphere.

III. Convergence Analysis of the MSCA Algorithm

Given nodes $i \in m$, and iteration t, the convergence process is divided into two sequential steps:
Step 1. The difference between γ_{t+1} and γ_t is bounded by that between (ξ^{t+1}, d_1^{t+1}) and (ξ^t, d_1^t); and
Step 2. (22) is non-increasing and lower-bounded.

A. Proof of Step 1

By proving with γ as an active variable, we introduce *Lemma* 1 as:

Lemma 1: Let $Y_{i,t+1} = 2[\sigma_{1,i,t}(\xi_{x,t+1} - \xi_{x,t}) + \sigma_{2,i,t}(\xi_{y,t+1} - \xi_{y,t}) + \sigma_{3,i,t}(\xi_{z,t+1} - \xi_{z,t}) + \sigma_{3,i,t}(d_{1,t+1} - d_{1,t})]$, with (28), $(\gamma_{t+1} - \gamma_t)^2$ is bounded by:

$$\begin{aligned}\left(\gamma_{t+1}-\gamma_t\right)^2 \le \frac{1}{\kappa^2}\left(\sigma_{4,i,t}-\sigma_{4,i,t-1}\right)^2+\kappa^2\left(Y_{i,t+1}-Y_{i,t}\right)^2+\kappa^2\left(\eta-1\right)^2\left[\left(\Phi_{i,t+1}-\Phi_{i,t}\right)-\left(\Phi_{i,t}-\Phi_{i,t-1}\right)\right]^2 \\ +4\kappa^2\left[\sigma_{3,i,t}\left(d_{i,t+1}-d_{i,t}\right)-\sigma_{3,i,t-1}\left(d_{i,t}-d_{i,t-1}\right)\right]^2=\varpi.\end{aligned} \tag{S1}$$

Proof: Given that (22) is non-convex, any equilibrium point with a zero-gradient (e.g., a local/global optimum or a saddle point) could be a feasible solution. Considering the solution to $d_{1,t}$ by (30b) as $d_{1,t+1}$, resulting in:

$$\gamma_t=\frac{\sum\sigma_{4,i,t}}{\sum\sigma_{3,i,t}}-\frac{N\mu\left(d_{i,t+1}-d_{i,t}\right)}{2\sum\sigma_{3,i,t}}-\frac{\kappa}{N}\sum\left(\Phi_{i,t}+Y_{i,t+1}\right). \tag{S2}$$

Inserting equations (30c) and (32c) into expression (S2), we can obtain:

$$\gamma_{t+1}=\frac{\sum\sigma_{4,i,t}}{\sum\sigma_{3,i,t}}-\frac{N\mu\left(d_{i,t+1}-d_{i,t}\right)}{2\sum\sigma_{3,i,t}}+\frac{\kappa\left(1-\eta\right)}{N}\sum\left(\Phi_{i,t+1}-\Phi_{i,t}\right)-\frac{\kappa}{N}\sum Y_{i,t+1}. \tag{S3}$$

From the aforementioned analysis, we can infer the difference between γ^{t+1} and γ^t as:

$$\begin{aligned}\gamma_{t+1}-\gamma_t=\left(\frac{\sum\sigma_{4,i,t}}{\sum\sigma_{3,i,t}}-\frac{\sum\sigma_{4,i,t-1}}{\sum\sigma_{3,i,t-1}}\right)-\frac{N\mu}{2}\left[\frac{\left(d_{i,t+1}-d_{i,t}\right)}{\sum\sigma_{3,i,t}}-\frac{\left(d_{i,t}-d_{i,t-1}\right)}{\sum\sigma_{3,i,t-1}}\right] \\ -\frac{\kappa\left(1-\eta\right)}{N}\left[\sum\left(\Phi_{i,t+1}-\Phi_{i,t}\right)-\sum\left(\Phi_{i,t}-\Phi_{i,t-1}\right)\right]-\frac{\kappa}{N}\sum\left(Y_{i,t+1}-Y_{i,t}\right).\end{aligned} \tag{S4}$$

With $(a - b - c - d)^2 \le 4(a^2 + b^2 + c^2 + d^2)$ being satisfied, we obtain:

$$\begin{aligned}\left(\gamma_{t+1}-\gamma_t\right)^2 \le \left(\frac{\sum\sigma_{4,i,t}}{\sum\sigma_{3,i,t}}-\frac{\sum\sigma_{4,i,t-1}}{\sum\sigma_{3,i,t-1}}\right)^2+\frac{N^2\mu^2}{4}\left[\frac{\left(d_{i,t+1}-d_{i,t}\right)}{\sum\sigma_{3,i,t}}-\frac{\left(d_{i,t}-d_{i,t-1}\right)}{\sum\sigma_{3,i,t-1}}\right]^2 \\ +\frac{\kappa^2\left(1-\eta\right)^2}{N^2}\left[\sum\left(\Phi_{i,t+1}-\Phi_{i,t}\right)-\sum\left(\Phi_{i,t}-\Phi_{i,t-1}\right)\right]^2+\frac{\kappa^2}{N^2}\left[\sum\left(Y_{i,t+1}-Y_{i,t}\right)\right]^2.\end{aligned} \tag{S5}$$

Based on (S5), *Lemma* 1 stands. □

B. Proof of Step 2

Lemma 2 is introduced to perform Step 2.

Lemma 2: Given the definition of the following intermediate variables, provided that the following criteria are satisfied:

$$\eta \ge 0, \kappa \ge 0, \tag{S6a}$$

$$\eta \le 1, \kappa \ge 0, \tau \ge 0, \mu \ge 0, \tag{S6b}$$

The following inequality stands:

$$L\left(\xi_{t+1},d_{1,t+1},\gamma_{t+1}\right)-L\left(\xi_t,d_{1,t},\gamma_t\right)\le 0, \tag{S7a}$$

$$L\left(\xi_{t+1},d_{1,t+1},\gamma_{t+1}\right)\ge 0. \tag{S7b}$$

Proof: Considering the second-order Taylor expansion and the optimal conditions of (30a) and (32a), the difference between $L(\xi_{t+1}, d_{1,t}, \gamma_t)$ and $L(\xi_t, d_{1,t}, \gamma_t)$ is given as:

$$L\left(\xi_{t+1},d_{1,t},\gamma_t\right)-L\left(\xi_t,d_{1,t},\gamma_t\right)\overset{(29a),(31a)}{=}\frac{4\kappa}{N}\sum_{i-1}^{N}\left[\sigma_{1,i,t}^2\left(\xi_{x,t+1}-\xi_{x,t}\right)^2+\sigma_{2,i,t}^2\left(\xi_{y,t+1}-\xi_{y,t}\right)^2+\sigma_{3,i,t}^2\left(\xi_{z,t+1}-\xi_{z,t}\right)^2\right]. \tag{S8}$$

Similarly, according to (30b) and (32b), the difference between $L(\xi_{t+1}, d_{1,t+1}, \gamma_t)$ and $L(\xi_{t+1}, d_{1,t}, \gamma_t)$ can be described by:

$$L\left(\xi_{t+1},d_{1,t+1},\gamma_t\right)-L\left(\xi_{t+1},d_{1,t},\gamma_t\right)\overset{(29b),(31b)}{=}\frac{4\kappa}{N}\sum_{i-1}^{N}\left[\sigma_{3,i,t}^2\left(d_{1,t+1}-d_{1,t}\right)^2\right]. \tag{S9}$$

Moreover, based on (30c), (32c) and (S1), the difference between $L(\xi_{t+1}, d_{1,t+1}, \gamma_{t+1})$ and $L(\xi_{t+1}, d_{1,t+1}, \gamma_t)$ is represented as:

$$L\left(\xi_{t+1},d_{1,t+1},\gamma_{t+1}\right)-L\left(\xi_{t+1},d_{1,t+1},\gamma_{t}\right)\overset{(28c),(30c)}{=}\frac{\left(\gamma_{t+1}-\gamma_{t}\right)^{2}}{\eta\kappa}\overset{S1}{\leq}\frac{\varpi}{\eta\kappa}. \tag{S10}$$

Combining (S7) with (S10), we can infer the following:

$$\begin{aligned}
&L\left(\xi_{t+1},d_{1,t+1},\gamma_{t+1}\right)-L\left(\xi_{t},d_{1,t},\gamma_{t}\right)\\
&\leq\frac{4\kappa}{N}\sum_{i-1}^{N}\left[\sigma_{1,i,t}^{2}\left(\xi_{x,t+1}-\xi_{x,t}\right)^{2}+\sigma_{2,i,t}^{2}\left(\xi_{y,t+1}-\xi_{y,t}\right)^{2}+\sigma_{3,i,t}^{2}\left(\xi_{z,t+1}-\xi_{z,t}\right)^{2}\right]\\
&+\frac{4\kappa}{N}\sum_{i-1}^{N}\left[\sigma_{3,i,t}^{2}\left(d_{1,t+1}-d_{1,t}\right)^{2}\right]\\
&+\frac{1}{\eta\kappa}\left(\frac{\sum\sigma_{4,i,t}}{\sum\sigma_{3,i,t}}-\frac{\sum\sigma_{4,i,t-1}}{\sum\sigma_{3,i,t-1}}\right)^{2}+\frac{N^{2}\mu^{2}}{4\eta\kappa}\left[\frac{\left(d_{i,t+1}-d_{i,t}\right)}{\sum\sigma_{3,i,t}}-\frac{\left(d_{i,t}-d_{i,t-1}\right)}{\sum\sigma_{3,i,t-1}}\right]^{2}\\
&+\frac{\kappa\left(1-\eta\right)^{2}}{\eta N^{2}}\left[\sum\left(\Phi_{i,t+1}-\Phi_{i,t}\right)-\sum\left(\Phi_{i,t}-\Phi_{i,t-1}\right)\right]^{2}+\frac{\kappa}{\eta N^{2}}\left[\sum\left(Y_{i,t+1}-Y_{i,t}\right)\right]^{2}.\\
&\Rightarrow\eta\geq0,\kappa\geq0.
\end{aligned} \tag{S11}$$

Therefore, based on the above conditions, (S6a) is satisfied, demonstrating that the augmented Lagrangian (22) does not increase. After (t+1)-*th* iteration, (22) is rewritten as:

$$\begin{aligned}
L\left(\xi_{t+1},d_{1,t+1},\gamma_{t+1}\right)=&\frac{1}{2N}\sum_{i=1}^{N}\left\|\hat{S}_{i}-S_{i}\left(O\right)\right\|_{2}^{2}+\frac{1}{N}\sum_{i=1}^{N}\left\langle\Phi_{i}\left(d_{1,t+1},\xi_{t+1}\right),\gamma_{t+1}\right\rangle+\frac{\kappa}{N}\sum_{i=1}^{N}\left[\Phi_{i}\left(d_{1,t+1},\xi_{t+1}\right)\right]^{2}\\
&+\frac{\mu}{2}\left[\left(d_{1,t+1}-d_{1,t}\right)^{2}+\left(\xi_{x,t+1}-\xi_{x,t}\right)^{2}+\left(\xi_{y,t+1}-\xi_{y,t}\right)^{2}+\left(\xi_{z,t+1}-\xi_{z,t}\right)^{2}\right].
\end{aligned} \tag{S12}$$

By substituting (S3) into (S12), we can obtain:

$$\begin{aligned}
L\left(\xi^{t+1},d_{1}^{t+1},\gamma^{t+1}\right)=&\frac{1}{2N}\sum\left\|\hat{S}_{i}-S_{i}\left(O\right)\right\|_{2}^{2}+\frac{\kappa}{N}\sum\left[\Phi_{i}\left(d_{1,t+1},\xi_{t+1}\right)\right]^{2}+\tau+\frac{\kappa\left(1-\eta\right)}{N^{2}}\left(\sum\Phi_{i,t+1}\right)^{2}\\
&+\frac{\mu}{2}\left[\left(d_{1,t+1}-d_{1,t}\right)^{2}+\left(\xi_{x,t+1}-\xi_{x,t}\right)^{2}+\left(\xi_{y,t+1}-\xi_{y,t}\right)^{2}+\left(\xi_{z,t+1}-\xi_{z,t}\right)^{2}\right].
\end{aligned} \tag{S13}$$

Let $\tau = (\sum\sigma_{4,i,}\sum\Phi_{i,t+1t}/N\sum\sigma_{3,i,t}) - \mu(d_{i,t+1}-d_{i,t})/(2\sum\sigma_{3,i,t})$ $-\kappa(1-\eta)\sum\Phi_{i,t+1}\sum\Phi_{i,t+1}/N$^2 $- \kappa\sum\Phi_{i,t+1}\sum Y_{i,t+1}/N$^2, with (S6), (S7) and (S13) is fulfilled, i.e., (22) is lower-bounded. Consequently, *Lemma* 2 stands, completing Step 2. □

In conclusion, based on the above deductions, the implementation of Steps 1 and 2 theoretically ensures the MSCA's convergence.